

A3 consolidation – renormalisation foundation (UV super-renormalisability + spectral continuum limit)

TECT verification-first repository · claim A3-UV-SUPERRENORMALISABILITY

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1. Purpose and scope

This is the single-place consolidation of the A3 renormalisation foundation — two operator-approved T6 theorems — and the entry note of the A3 reproduction bundle. It establishes that the scalar Brazovskii functional in $d = 3$ is ultraviolet-finite and that its perturbative correlators have a continuum limit (spectral regularisation). Full proofs: [claims/A3-UV-SUPERRENORMALISABILITY/notes/a3-uv-superrenormalisability-260623-260623-v1.1.tex.txt](#) and [claims/A3-PERTURBATIVE-CONTINUUM-CORRELATORS/notes/a3-graphwise-convergence-lemma-260623-260623-v1.4.tex.txt](#). Numerics: [codes/foundations/a3_renormalisation_checks.py](#) (6/6) and [codes/foundations/a3_graphwise_convergence_checks.py](#) (8/8). Scope: scalar, $d = 3$, $\mu^2 > 0$, $\gamma > 0$, perturbative order-by-order.

2. The two theorems

Theorem A (UV super-renormalisability, A3-UV). *With the q^{-4} propagator $G = 1/K$, every connected diagram in $d = 3$ with $V \geq 1, I \geq 1$ has $D = 3 - 3V - I < 0$; by Weinberg’s theorem the perturbative expansion is UV- and IR-finite. No UV-divergent counterterm is required; a finite normal ordering is an optional scheme choice.*

Theorem B (spectral continuum limit, A3-CORRELATORS). *With the spectral/Galerkin regulator $G_a(q) = \mathbf{1}_{\{|q| \leq \pi/a\}}/K(q)$ and $\gamma > 0$ (measure well-defined), every connected perturbative amplitude satisfies, for each FIXED external momentum p , $\lim_{a \rightarrow 0} \mathcal{A}_{G,a}(p) = \mathcal{A}_G(p)$; hence each order of the perturbative correlators has a cutoff-independent continuum limit ($\Lambda = \pi/a$, so $a \rightarrow 0 \equiv \Lambda \rightarrow \infty$).*

3. Proof structure

Theorem A: power counting $D = (d-4)I - dV + d < 0$ for $d \leq 3$; all subgraphs satisfy $D' < 0$, so Weinberg gives absolute convergence (UV-finite). Theorem B: the spectral bound $G_a(q) \leq 1/K(q) \leq C(1 + |q|)^{-4}$ ($C = 1.6 \times 10^3$, a -independent) holds for ALL unwrapped internal-line momenta; with Theorem A’s Weinberg integrability the dominating function $\prod_i C(1 + |q_i|)^{-4} \in L^1$, so dominated convergence gives the fixed- p limit. $\gamma > 0$ makes $F_{\Lambda,a}$ bounded below ($Z_{\Lambda,a} < \infty$), defining the measure.

4. Recorded flaw and open refinement

The finite-difference lattice regulator $\hat{q}_a = \frac{2}{a} \sin \frac{aq}{2}$ was *refuted* for the domination: $\hat{q}_a$ is periodic, so an unwrapped internal-line momentum exceeding the BZ folds ($\hat{q}_a \approx 0$ at $q \sim 2\pi/a$) and G_a^{latt} is not bounded by $C(1 + |q|)^{-4}$ (aliasing/Umklapp; [lattice_folding_flaw_demonstrated](#)). Theorem B therefore uses

the spectral regulator (Route A). The genuine finite-difference lattice continuum limit (and regularisation-independence) needs a Reisz-type lattice power-counting theorem — **Route B, OPEN**. Uniformity on compact p -sets is also not claimed.

5. Devil's-advocate (closed in review)

(α) UV power counting is only superficial — Weinberg (all $D' < 0$) gives absolute convergence. (β) UV-finiteness $\neq$ continuum limit — hence the split; Theorem B proves the latter separately. (γ) the lattice aliasing flaw — recorded (§4) and avoided by the spectral regulator. (δ) uniform-on-compact- p — not claimed (per- p DCT only). Quantitative sanity check: $D < 0$ table; spectral $C = 1.6 \times 10^3$ (a -independent); lattice folding ratio $\sim 10^7$ – 10^{11} (the flaw); Weinberg integrability; $\Lambda = \pi/a$.

6. Result footer

Result ID:	A3 consolidation (UV + spectral continuum limit)
Precise statement:	(A) $d=3$ q^{-4} Brazovskii is UV-finite ($D=3-3V-I<0$, Weinberg); no UV-divergent counterterm. (B) spectral regulator $\Rightarrow$ each connected perturbative amplitude converges for each fixed p ; perturbative correlators have a cutoff-independent continuum limit.
Scope:	scalar, $d=3$, $\mu^2>0$, $\gamma>0$, perturbative; spectral regulator; fixed external momenta.
Dependencies:	A1-KERNEL-CONV.
Hypotheses:	A3-H1-DIM3-Q4-KERNEL; A3-H2-IR-POSITIVITY; A2-H2-SEXTIC-COERCIVITY ($\gamma>0$, measure).
Evidence grade:	PROVED CONDITIONAL (analytic) + EXECUTED (6/6 + 8/8).
Reproduction command:	python codes/foundations/a3_renormalisation_checks.py && python codes/foundations/ a3_graphwise_convergence_checks.py
Expected output:	6/6 and 8/8 PASS; exit 0.
Tier:	T6 PROVED CONDITIONAL (operator-approved 2026-06-23).
Open refinement:	genuine finite-difference lattice (Reisz) = Route B.
No-overclaim statement:	perturbative order-by-order; spectral (NOT finite-difference lattice); fixed- p (NOT uniform-on-compact); NOT constructive; NOT Class-II; NOT $\mu^2<0$.
Next required action:	(bundle) referee reproduction; Route B / A3_constructive are separate targets.